

Western Digital
Media Centre

lot—that honour goes to Maxtor. The drives from Maxtor have the strongest cases we've ever seen. They're built of solid metal! You can keep stacking and these babies will take all the weight you can throw at them, and more!

The drives that disappointed here were the WDs, which were rather plain-looking, and had bodies of plastic that felt flimsy. If you were to try stacking up these drives, we wouldn't be surprised to see some cracks after a while!

All the drives—except for WDs—in the desktop storage/backup category came with a single button. This, when pressed once, activates the power; when pressed twice, it acts as a 'one-touch' backup button. If you keep it pressed for a while, the drive shuts down.

The WD drives came with three buttons: one dedicated power button, with the other two for automatic backup and on-demand

backup. In automatic backup, the drive just goes ahead and backs up stuff pre-specified by the user. In on-demand backup, the backup software opens up and you can back up data as you please.

The only drive that featured something apart from what we've mentioned so far was the Western Digital Media Center. It featured a Multiple Multimedia Card Reader, which is stipulated to read pretty much every kind of card you can throw at it. You can copy stuff to the hard disk directly using the on-demand backup button. There is no screen, so you're limited to copying the entire contents of the card—you can't see or choose what you're copying.

On To The Tests

Synthetic Benchmarking: The test process used here was exactly the same as with the portable drives.

Almost every drive we got here was a 7,200 rpm drive. For some reason, the Maxtor 250 GB was a mere 5,400 rpm drive with an extremely low 2 MB of cache. All the others had at least 8 MB, and the Maxtor One Touch II came with 16 MB of cache onboard.

We started with the synthetic benchmarking using SiSoft Sandra 2005. One drive that absolutely blew everyone away was the Maxtor One Touch II. It



Jargon buster

ATA: Advanced Technology Attachment. This is a standard interface for connecting storage devices such as hard drives, CD-ROMs, etc. It allows only 18 to 36 inches of cable length, and is therefore seen mostly as an internal solution.

Cache: This is a buffer available to the hard drive, and is usually a physically separate entity on the hard drive's IC. It synchronises data transfers between the disk, reading it off the platter and the disk's I/O system to the computer. Normally, a larger cache will result in faster read/write speeds and access times.

DMA: This gives devices within the computer the ability to communicate to and access the system memory without utilising the CPU—which results in faster overall computing. All modern disk drives support this.

Heads and Actuators: Modern hard disks have multiple heads and multiple platters to read from. The head is what reads the data from the platter, and the actuator is the arm that moves the head across the areas of the platter.

IDE/EIDE: IDE stands for Integrated Drive Electronics, and is the standard used for all the drives commonly found in computer systems today. EIDE is enhanced IDE; it is a specific type of attachment interface specification that allows for high-performance, large-capacity drives.

MTBF: Mean Time Between Failures. This is essentially the reliability index signifying the average time for which the device will definitely work. Typically, hard drives have an MTBF rating of 500,000 hours.

Head Parking: When the platter of the hard drive is not spinning, the head rests on the surface of the disk. When the platter does start spinning, the head drags along the surface till enough speed is gathered for it to disengage from the surface and stay nanometres above the platter to read or write data. This dragging can corrupt data, so there is a special track where no data is written and is used for the head while it is idle or gathering speed.

Platter: The flat disk that actually holds all the data on the hard disk, in magnetic format. When requested, the head reads the data off them.

SCSI: Small Computer Systems Interface; pronounced 'skuzzy'. This is an advancement of the IDE/ATA interfaces. While IDE is an interface, SCSI effectively is a system-level bus with controllers on each SCSI device working to optimise data flow. SCSI has multiple standards such as ANSI, ITIC, NCITS and T10.

SATA: Serial ATA. It was primarily developed to replace the parallel ATA technology. Transfer speeds for SATA start at 150 Mb/s, and the newer SATA II is touted as being capable of 3 Gb/s.

Spindle Speed (rpm): This is the speed at which the platter in the hard drive spins. The faster it spins, the faster the access time, as the time taken by the head to reach and access the data is lesser. Till recently, 5,400 rpm was quite common, but with the fall in prices, more and more computers are shipping with 7,200 rpm drives. Laptops usually have 4,200 rpm drives.

Seek Time: The seek time is the amount of time between when the CPU requests a file and when the file is sent. On an average, seek times are in the 2 to 5 ms range for internal hard drives. It's a lot more for external hard drives due to the interface limitations.

1/4th page Ad